

Heat & Thermal Energy

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Heat always moves from **warmer** objects to **cooler** ones until they are the same temperature. Good **conductors** (a metal spoon) let heat move quickly. **Insulators** (wood, a foam cup, an oven mitt) slow heat down.

Round 1 — Retrieve It

1. Which way does heat move — from warmer to cooler, or from cooler to warmer?

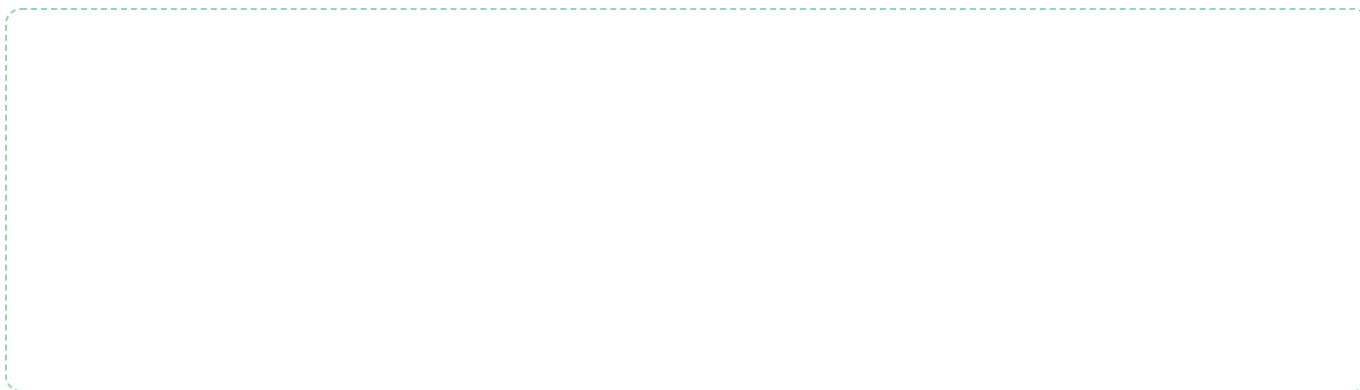
2. Heat keeps moving between two objects until what happens?

3. What do we call a material that lets heat move through it quickly?

4. Name one material that is a good insulator.

Round 2 — Sketch It

Draw a metal spoon sitting in a cup of hot cocoa. Use an arrow to show which way the heat moves. Label the warmer part and the cooler part.



Heat & Thermal Energy — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. You hold an ice cube in your warm hand and it starts to melt. Which way is the heat moving, and from what to what?

2. Why do cooks use an oven mitt instead of a metal fork to take a hot pan out of the oven?

3. Hot soup is served in a foam cup instead of a metal cup so it stays warm and is safe to hold. Explain why the foam cup works better.

Round 4 — Reason from Evidence

A student puts one metal spoon and one wooden spoon into the same cup of hot water. After two minutes, the handle of the metal spoon feels hot but the wooden spoon handle feels only a little warm. What does this show about metal and wood?

Heat & Thermal Energy — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain the rule for how heat moves between a warm object and a cool object, and tell when the heat stops moving.

C**Connect**

Connect the idea of insulators to something in your home that keeps things hot or cold. Name it and explain how it works.

E**Extend**

Plan a test to find out which cup keeps warm water warm the longest. Tell what you would measure and how.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Heat & Thermal Energy

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. From warmer to cooler.
2. Until both objects are the same temperature.
3. A conductor (a good heat conductor).
4. Any: wood, foam, plastic, or an oven mitt (cloth).

Sketch It — look for

An arrow showing heat moving from the hot cocoa (warmer) into the metal spoon (cooler), with the warmer and cooler parts labeled.

Apply It

1. Heat moves from your warm hand to the cooler ice cube, which warms and melts the ice.
2. A metal fork is a conductor and would let heat move quickly to their hand; an oven mitt is an insulator that slows the heat, so it protects the hand.
3. Foam is an insulator, so it slows the heat from leaving the soup and slows heat from reaching your hand, keeping the soup warm and safe to hold.

Reason from Evidence — look for

Metal is a good conductor, so heat moved quickly up the handle and it felt hot. Wood is an insulator, so it slowed the heat and the handle stayed cooler.

ACE — Extend look-for

Plan pours the same amount of warm water into different cups and uses a thermometer to measure the temperature (in °C) over time; the cup whose water stays warm longest is the best insulator.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.